How the System Functions

An end-to-end functional overview of the AI-assisted recruitment platform: how candidates, hiring managers, and interviewers move through screening, scheduling, interviewing, and evidence-backed hiring decisions.

Functional & Architecture Overview · Generated 26 October 2026

1. What RecruitO Does

Recruitment teams receive far more applications than they can assess consistently. Screening is shallow, interviewers apply different standards, and interview content is lost in unstructured notes. **RecruitO** supports the hiring workflow from application intake to a consolidated hiring decision, and every AI-generated score or flag is explainable and traceable to CV content or a transcript segment.

The guiding principle is that RecruitO **augments human decision-making, it does not replace it**. Scores support human review; hiring decisions remain with people.

The three stages of the hiring journey

Stage	What RecruitO does
Pre-interview	Accepts PDF/DOCX applications and hiring-manager batch imports of up to 50 CVs. Every CV is scored automatically with quoted evidence per requirement and ranked per job.
During interview	Emails the hiring manager when a candidate is interview-eligible. The manager assigns the date, time, and interviewer; RecruitO emails the candidate and interviewer the confirmed details, a calendar attachment, and expiring role-scoped links to a browser interview room.
Post-interview	With participant consent, captures transcript segments, analyses interview evidence, and produces printable consolidated reports and candidate comparisons.

2. Who Uses It (Roles)

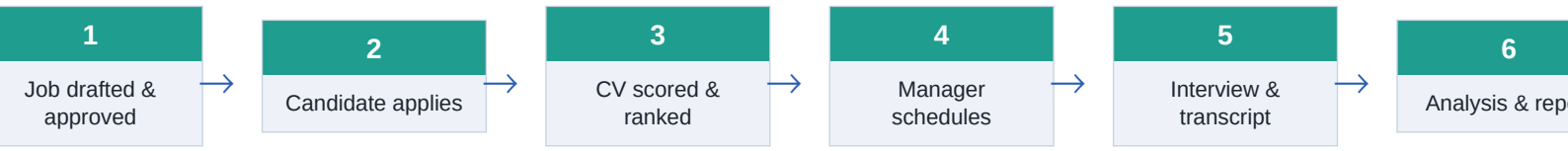
RecruitO is role-aware. Each role sees only its own data, enforced by per-role allowlists and user-bound database access.

Role	What they do
Hiring Manager	Drafts and approves job descriptions with the AI assistant, reviews candidate evidence and rankings, imports CV batches, assigns interview date/time and an interviewer, publishes jobs to Blogger, and makes the final hiring decision.
Interviewer	Receives assigned interviews and transcripts, joins the interview room via a role-scoped link, and contributes to structured assessment.
Candidate	Creates an account, submits the application form with a CV, receives the interview date/time and join link by email, gives (or revokes) consent for transcription and biometric matching, and tracks status in their dashboard.

A role-aware chatbot answers questions using only the signed-in user's own status and public job cards.

3. The End-to-End Flow

At a glance, an application travels through these steps:



1. Job preparation. A hiring manager gives the role title, seniority, key skills, employment type and location mode. The JD agent retrieves stored hiring policies, drafts the description, extracts structured requirements, checks the draft against policy (retrying if needed), and saves it for human review and approval.

2. Application. The manager shares the public application link. A candidate creates an account and submits the form with a PDF or DOCX CV. Managers can also batch-import up to 50 CVs for an approved job; each file is validated and stored independently, so an unreadable or duplicate CV is reported without stopping the batch.

3. Screening & scoring. Each CV is queued for assessment: text is extracted, every requirement is evaluated with quoted evidence verified against the CV text, category scores and justifications are built, and applicants are ranked by total score with a skills-then-experience tie-breaker. Scoring is deterministic (temperature 0) and cached by CV hash, requirements version, prompt version and model.

4. Scheduling. A completed latest score strictly above 65/100 makes a portal candidate eligible for an interview (lower scores are not auto-rejected). RecruitO emails the hiring manager that the candidate is ready, with a link into the hiring workspace. The manager opens the application and assigns an interview date and time (a standard browser date-and-time picker) and an interviewer from a dropdown of active interviewers. There is no external calendar integration and no OAuth to configure — the

manager is the scheduling authority.

5. Interview. RecruitO emails the candidate and interviewer the confirmed date and time, an `.ics` calendar attachment, an “add to calendar” link, and their role-scoped join links. Candidate and interviewer join a browser WebRTC room via those links. With explicit, separate consent, short (~3.5 s) audio chunks are transcribed and the transcript text is stored; an optional biometric opt-in stores only a face embedding that expires after 30 days.

6. Analysis & decision. The analysis graph produces structured Q&A, maps answers to requirements with transcript references, and flags unexplored requirements. The manager opens a printable consolidated assessment, compares candidates, and makes the final decision.

4. How It Is Built (Architecture)

The browser frontend calls a FastAPI backend. PostgreSQL stores job, candidate, interview, score, audit, notification and consent records. LangGraph coordinates the multi-step AI workflows, Google Gemini provides the language and audio intelligence, and OpenCV runs face matching locally inside the API container. The whole stack starts with one Docker Compose command.

Technology stack

Layer	Technology	Role
Backend	Python, FastAPI, Pydantic	Typed request/response models and automatic OpenAPI docs.
AI orchestration	LangGraph	Multi-step, stateful hiring workflows.
Database	PostgreSQL (pgvector)	Relational integrity for all hiring records.
Frontend	HTML / CSS / JS, Nginx	Small static frontend served in Compose.
Video	WebRTC + FastAPI WebSocket	Peer media in the browser; backend relays signalling.
Speech-to-text	Gemini audio	Consented chunked transcription (not low-latency streaming).
Face matching	OpenCV YuNet / SFace	Optional opt-in; embeddings run and expire locally.
LLM provider	Google Gemini API	JD drafting, CV scoring, chat, transcription, assessment.
Email	SMTP (Gmail)	Application, status, interview and reminder emails.
Containers	Docker, Docker Compose	One-command reproducible startup.

LangGraph workflows

Graph	Purpose
JD graph	Retrieve policies, draft the JD, extract requirements, check policy, retry, persist.
Candidate chatbot	Answer using the signed-in candidate's own status and public job cards.
Role assistant	Per-role, user-bound tools for manager / interviewer / candidate data.
CV assessment	Load inputs → extract text → evaluate → persist, re-running up to three times if output is malformed or quotes evidence absent from the CV.
Application ranking	Rank scored applicants by total score, then skills and experience points.
Interview analysis	Produce F5 analysis and persist, then conditionally evaluate deeper dimensions.

5. Notifications & Email Delivery

RecruitO queues outbound email (application receipts, status changes, interview offers and confirmations, and interviewer reminders) and a background worker dispatches it. Email is delivered through the configured **SMTP server**: when `SMTP_HOST` is set, the worker uses the real `SMTP_*` credentials; the `APP_MAIL_*` values are only a fallback used when no SMTP host is configured.

- An interview-eligibility email goes to the hiring manager only for portal applicants whose latest CV score is above 65/100.
- Once the manager schedules, RecruitO emails the candidate and interviewer the confirmed details, an `.ics` attachment and an “add to calendar” link, plus an interviewer reminder — all over email, with no external calendar service.
- Batch-imported CVs use reserved `.invalid` addresses and are never emailed.

Note: because delivery now uses a live SMTP server, messages reach real inboxes — use test recipient addresses when demonstrating.

6. Privacy & Data Handling

- **Data minimisation.** Only transcript text is stored from audio; source chunks are not retained. Biometric opt-in stores only an SFace embedding for up to 30 days; photos and frames are processed in memory and discarded.
- **Consent is explicit and separable.** Transcription and biometric matching are independent choices, and consent can be revoked.
- **Third-party disclosure.** CV text, consented audio, and assessment prompts go to Google Gemini; face matching runs locally; interview times are entered by the manager and shared only over email, with no external calendar service.
- **Deletion.** Managers can delete a candidate's data (with a reason); applications, scores, transcripts, consents and the embedding cascade, and personal values in earlier audit entries are cleared.
- **Fairness.** Scoring compares CV text to stated requirements and supports human review; it does not make automated rejection decisions.

7. Running the System

The entire stack is containerised. From the project root:

Step	Command
Build & start everything	docker compose up --build
Seed the hiring-manager account (first run)	docker compose exec api python -m app.scripts.seed_staff
Stop (data kept)	docker compose down

Once running:

Surface	URL
RecruitO frontend	http://localhost:5500
API health check	http://localhost:8000/health
Interactive API docs (OpenAPI)	http://localhost:8000/docs

All assessment output is decision support. RecruitO makes the hiring process faster and more consistent; people make the hiring decisions.